

Types of Testing Techniques

Software testing techniques are methods used to check whether a system works as expected and meets its requirements. Below are some commonly used testing techniques:

1.Black Box Testing

This technique tests the system without looking at the internal code. It focuses on inputs and expected outputs to check if the system behaves correctly. It's useful for testing user interfaces, forms, and overall system functionality.

2.White Box Testing

In white box testing, the internal logic and structure of the code are tested. It involves checking loops, conditions, and paths in the program. This type of testing is usually done by developers.

3.Unit Testing

This involves testing individual components or functions of the software in isolation. For example, checking if the login function works correctly by itself.

4.Integration Testing

Integration testing checks if different modules or parts of the system work together correctly. For example, ensuring that after login, the user is correctly directed to their dashboard.

5.System Testing

System testing involves testing the complete system as a whole to verify that it meets all specified requirements. It covers both functional and non-functional aspects.

6.Acceptance Testing

Also known as User Acceptance Testing (UAT), this is done to verify if the system is ready for end-users. It checks whether the software meets business needs and user expectations.

7.Regression Testing

This testing ensures that new changes or updates have not broken any existing functionality. It involves re-running previous test cases after updates or bug fixes.

8.Performance Testing

This checks how well the system performs under load — for example, how fast it loads pages or handles multiple users at once.

These testing techniques help ensure software quality, reliability, and user satisfaction.